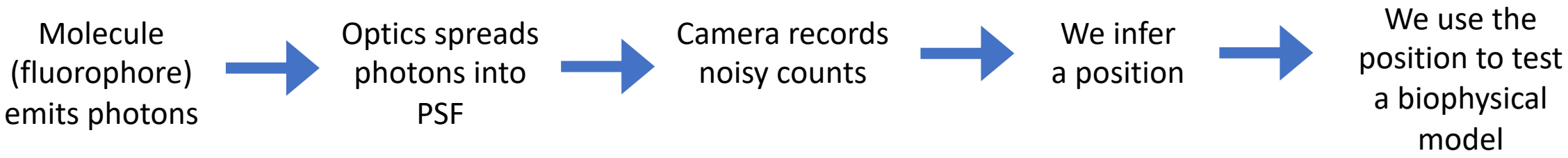
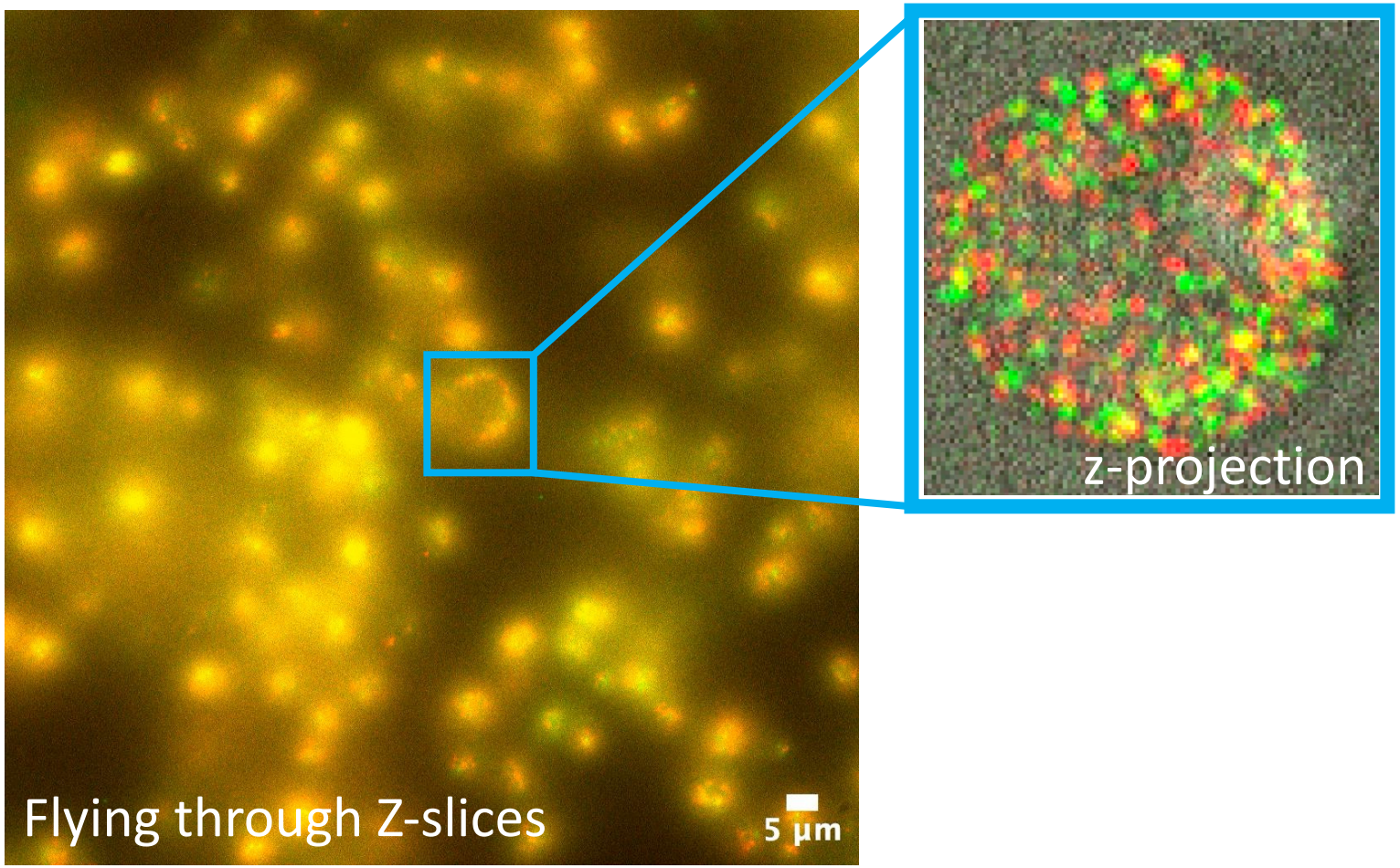
Main idea:


Molecule (fluorophore) emits photons
Optics spreads photons into PSF
Camera records noisy counts
We infer a position
We use the position to test a biophysical model

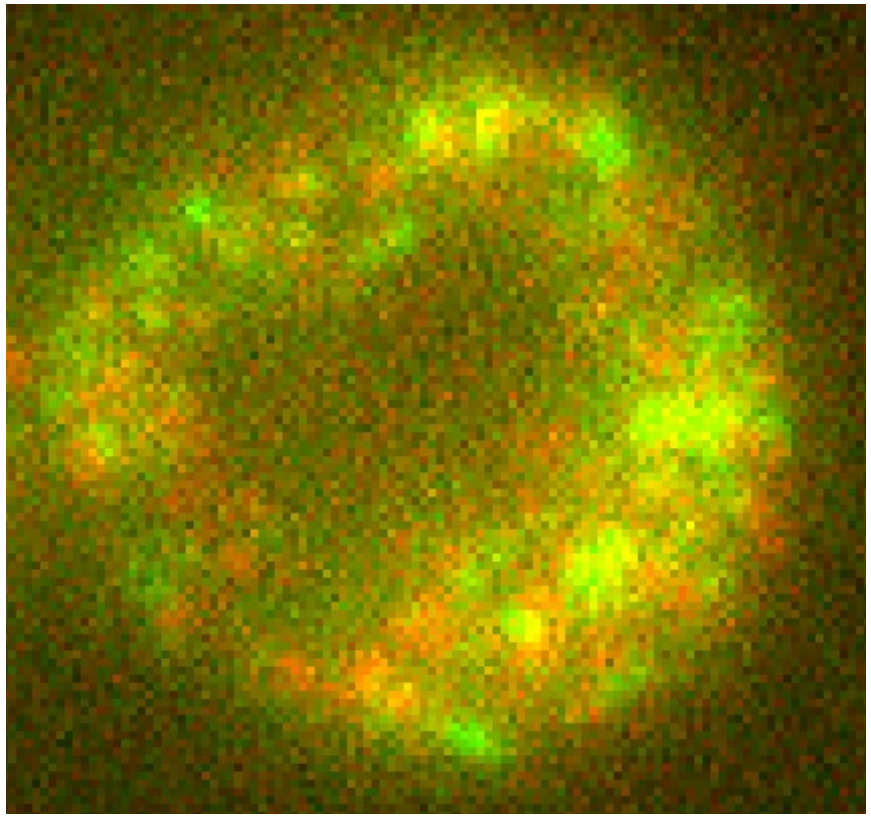
3D microscopy example: Single-molecule RNA FISH



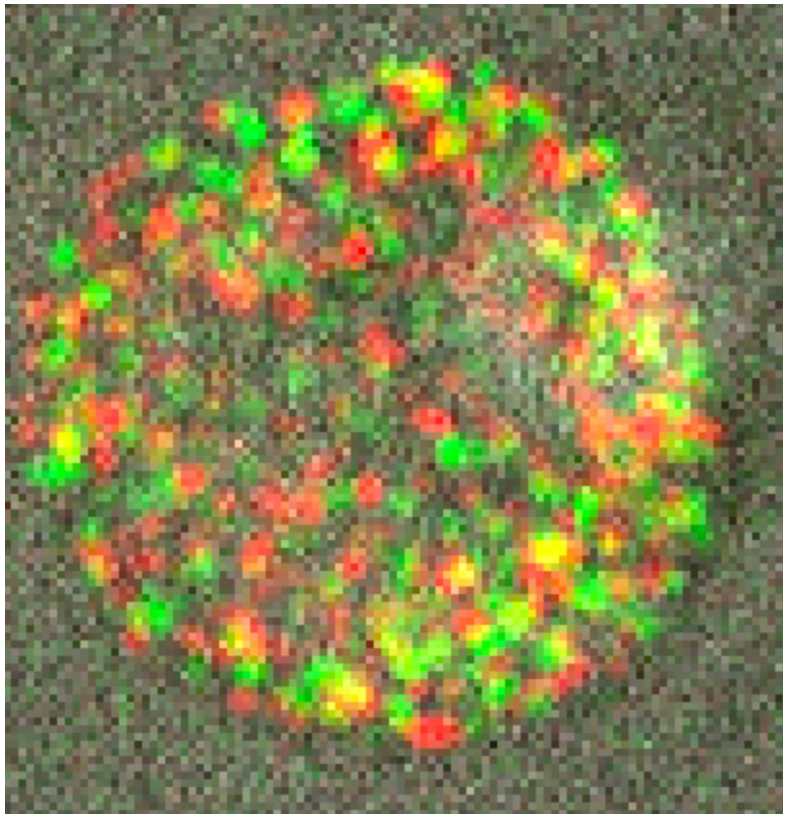
RPL3 mRNA: ribosomal protein of large 60S
RPS3 mRNA: ribosomal protein of small 40S

Laman Trip et al.
Nature Communications (2022)

A fluorescent spot is an optical image



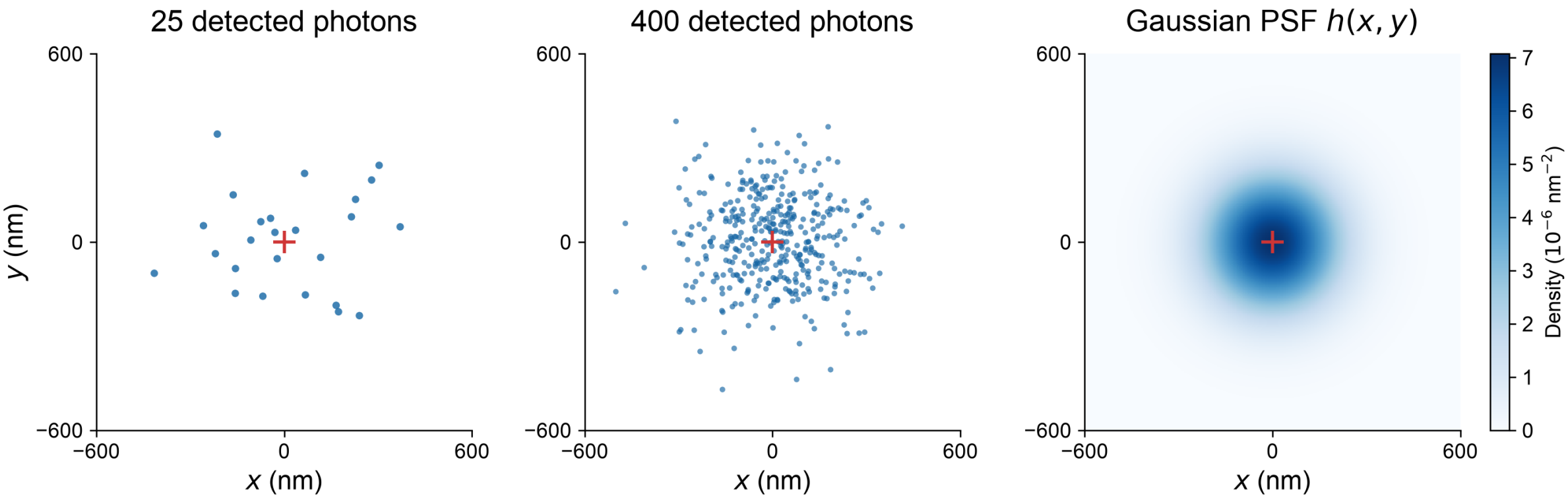
One optical slice



42 slices projected onto xy-plane

The same cell. What sets the width of each bright spot?

One stationary molecule → many photon positions



Simulated photon positions. Red crosses mark the true emitter position. $s = 150$ nm.

A single frame samples this distribution (PSF) with a finite number of photons.

The normalized point-spread function

P = probability of a photon being detected at r, given fluorophore at r_0

$$h(\mathbf{r}-\mathbf{r}_0)\,d^2r = P(\text{landing in } d^2r \mid \mathbf{r}_0)$$

$$\int h(\mathbf{r})\,d^2r = 1$$

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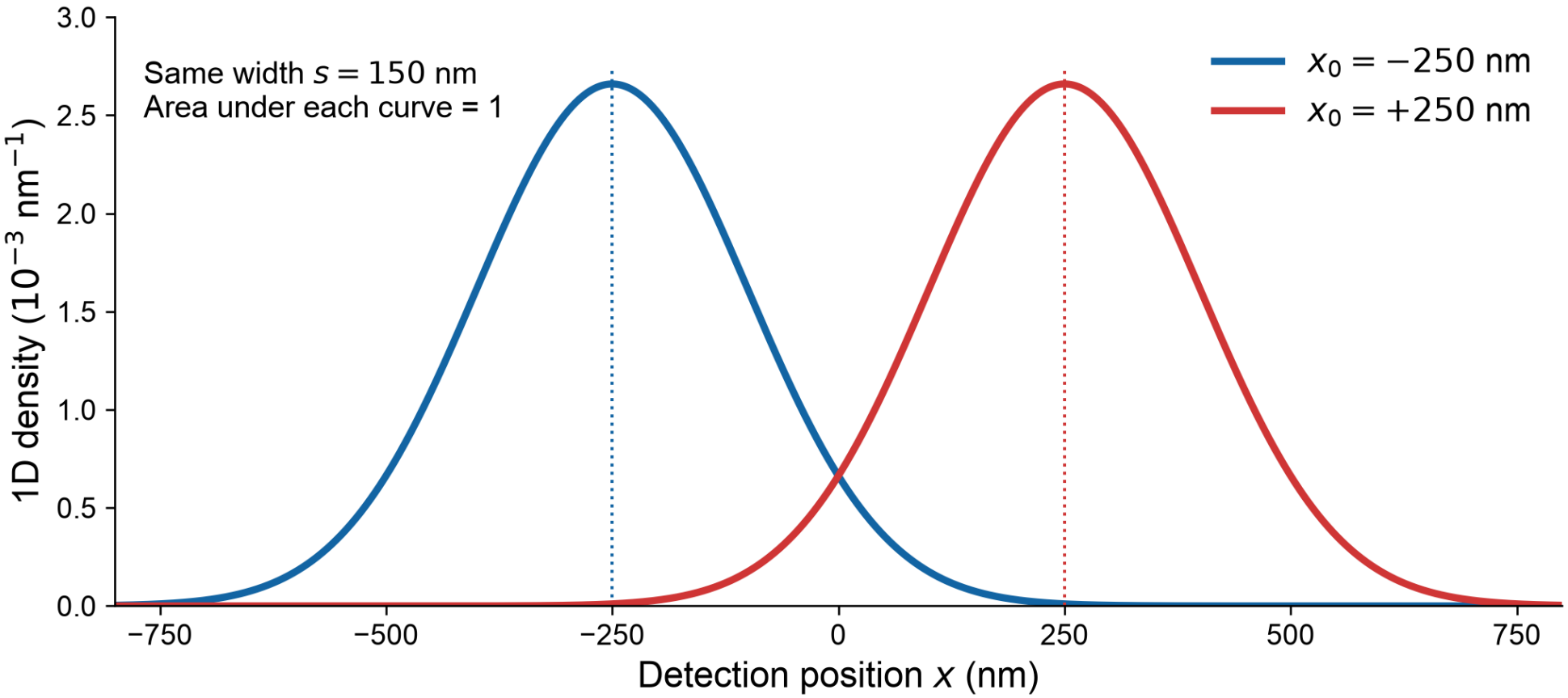
h: spatial probability density

q: detected counts per second

$$\lambda_s(\mathbf{r},t) = q(t)\,h(\mathbf{r}-\mathbf{r}_0)$$

λ_s is a detected count rate per unit area. The PSF redistributes light over position.

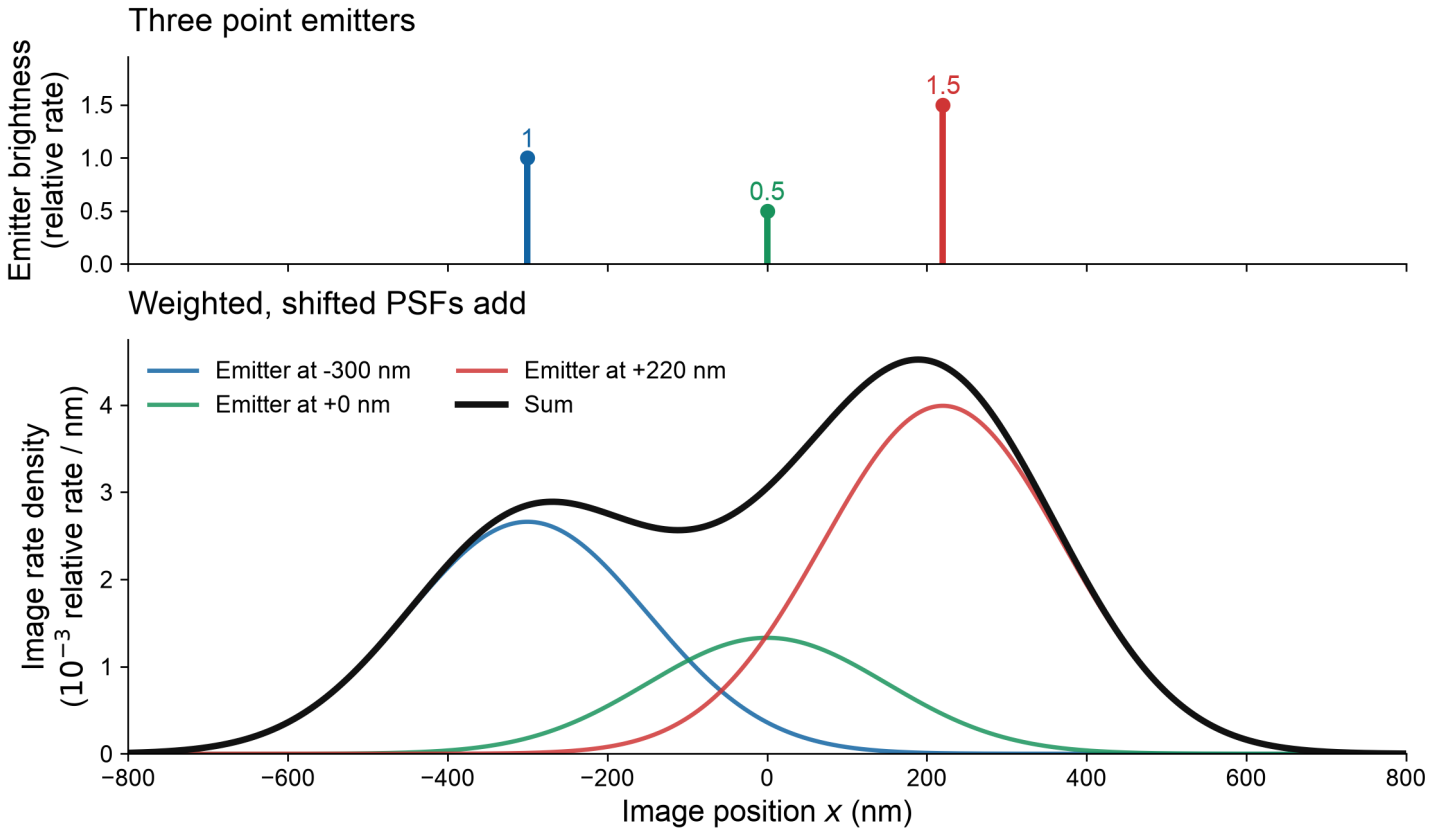
Moving the source (fluorophore) shifts its image



Source position r′ → image contribution h(r − r′)

Shift invariance is an approximation over a limited field and depth range.

Images of independent emitters add



Place a PSF at each source.

Scale by its brightness.

Add the contributions.

$$\lambda_s(\mathbf{r}, t) = \sum_j q_j(t)\, h(\mathbf{r} - \mathbf{r}_j)$$

Illustration in one dimension. The same construction works in the image plane (i.e., on “focal plane” = xy-plane).

Convolution adds all source contributions

Source patch: ΔA′ near r′ Target patch: ΔA near r

Rate into the target patch = source rate × landing probability

$$f(\mathbf{r}', t)\,\Delta A' \times h(\mathbf{r} - \mathbf{r}')\,\Delta A$$

Sum the source patches, then divide by the target area ΔA.

$$\lambda_s(\mathbf{r}, t) = \int h(\mathbf{r} - \mathbf{r}')\, f(\mathbf{r}', t)\, d^2 r'$$

f: detected source rate per area. Hold r fixed and integrate over source positions r′.

Exposure time converts rates into counts

Stationary emitter: mean signal in pixel i = qT p_i = Np_i

$$\mu_{i,n} = \int_{t_n}^{t_n+T} dt \int_{A_i} d^2 r \left[\lambda_s(\mathbf{r},t) + \lambda_b(\mathbf{r},t)\right] + d_i T$$

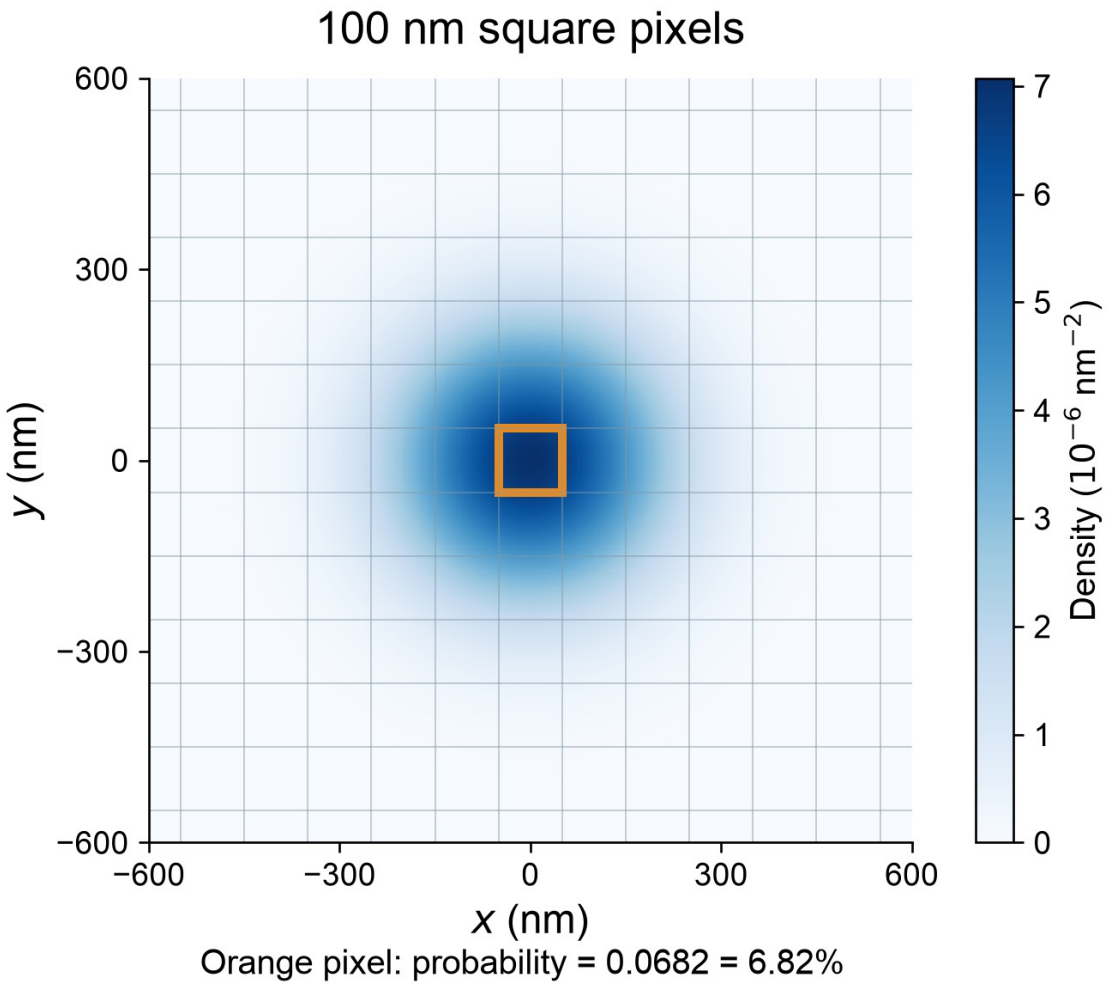
Integrate over pixel area

Integrate over exposure

Add dark-current charge

This gives the expected count $\mu_{i,n}$. It does not yet tell us its fluctuations.

Each pixel collects a fraction of the PSF



$$p_i(\mathbf{r}_0) = \int_{A_i} h(\mathbf{r} - \mathbf{r}_0)\, d^2 r$$

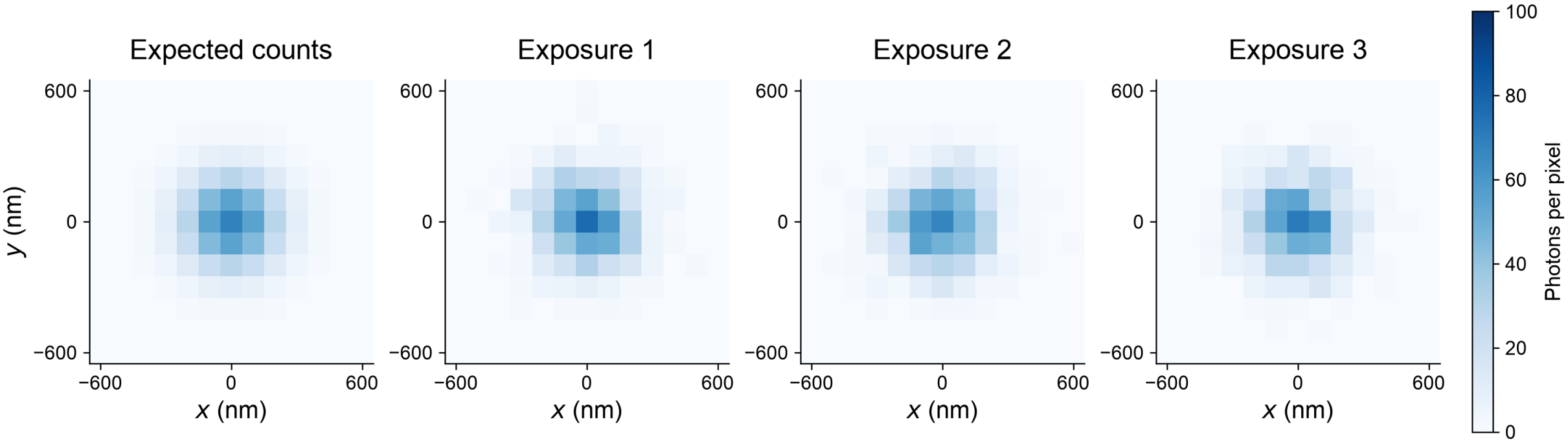
$$\mu_i = N p_i$$

Gaussian width: s = 150 nm
Pixel width: a = 100 nm

Central-pixel probability: 0.0682

1,000 expected total
→ 68.2 in this pixel

Expected counts and repeated measurements



Same stationary emitter and exposure conditions. $N = 1000$, $s = 150$ nm, $a = 100$ nm.

The specimen and illumination stay fixed. The measured counts still vary.

Simulated measurements; shot noise only. Every panel uses the same count scale.

Photon counts and camera readout

Expected count

μ_i

A mean over repeats

→

Accumulated charge

Z_i

An integer in one exposure

→

Measured readout

Y_i

Includes electronic noise

$$Z_i \sim \text{Poisson}(\mu_i), \qquad Y_i = Z_i + \varepsilon_i$$

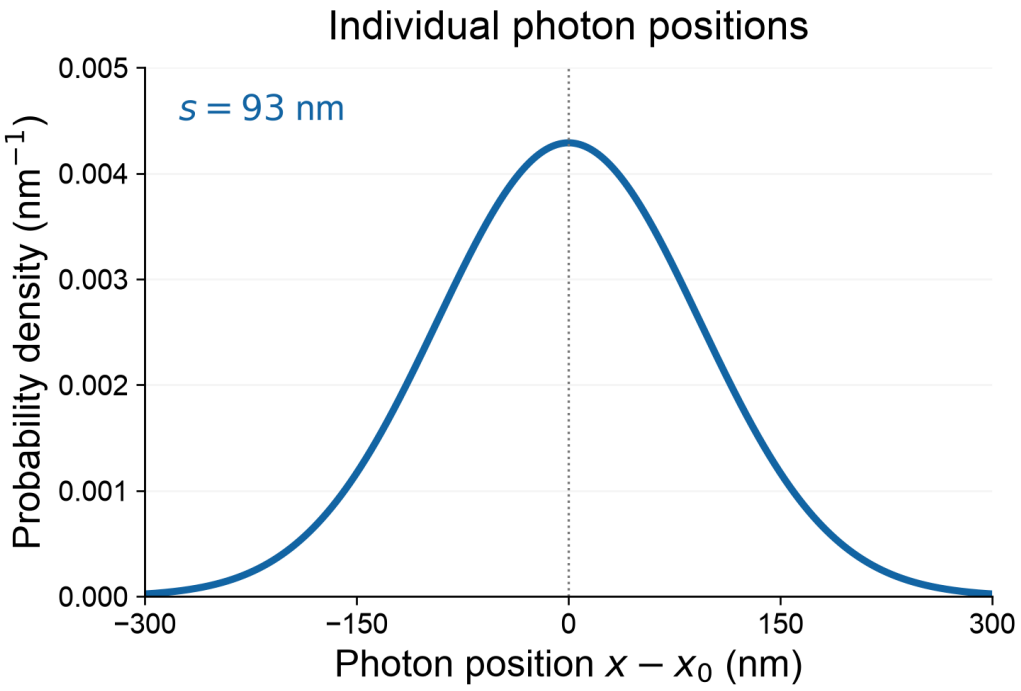
$$\varepsilon_i \sim \mathcal{N}(0, \sigma_{r,i}^2)$$

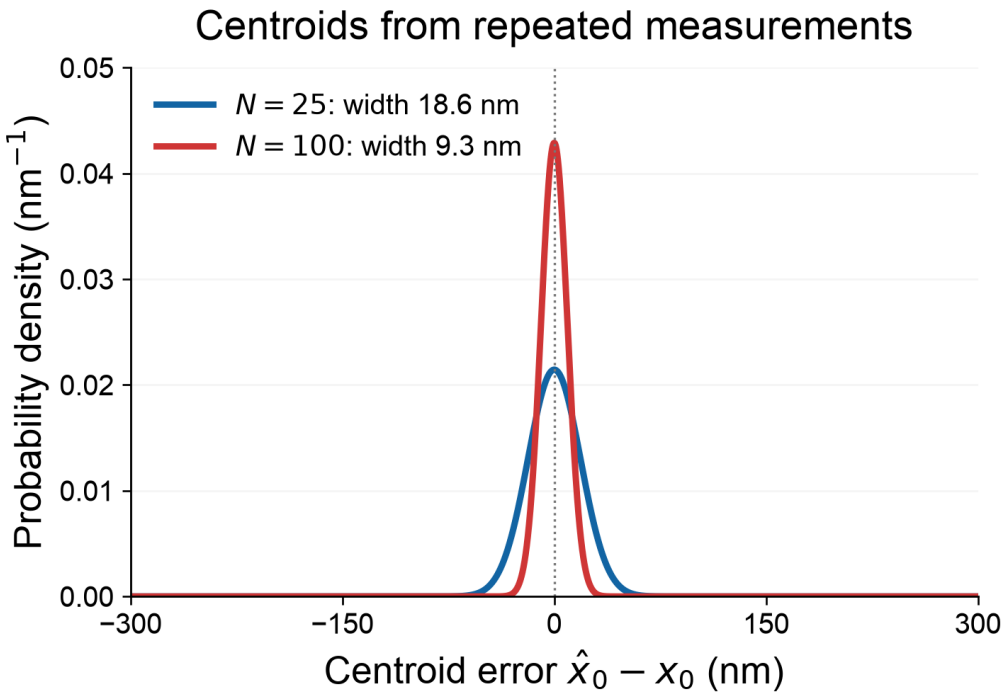
Raw camera values: $C_i = o_i + g_i Y_i$

Calibrate offset o_i and gain g_i before interpreting digital values as electron counts.

One broad spot can yield a precise position

The PSF describes photon landings. Repeated detections constrain its center.



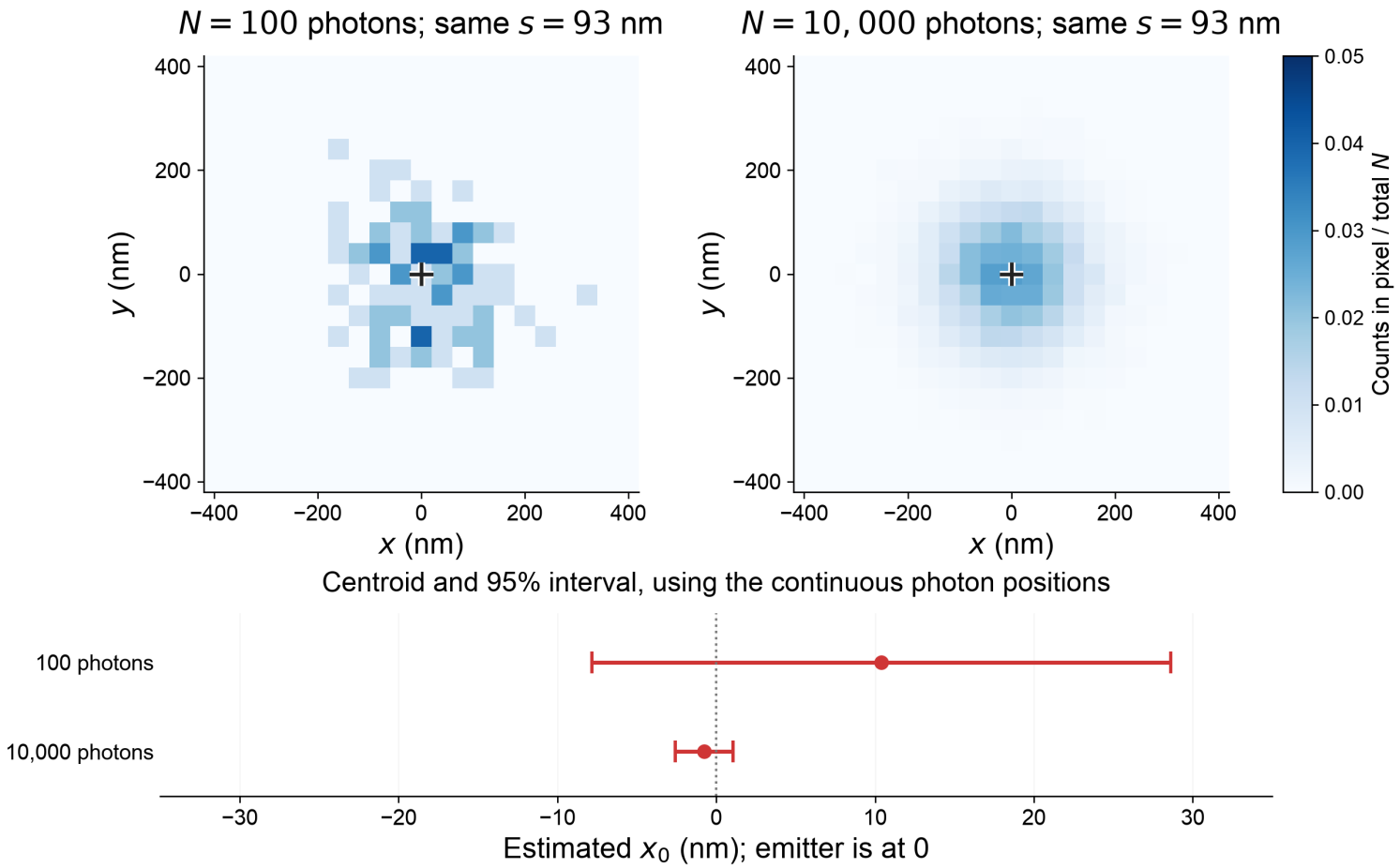


s: width of the photon distribution

s/√N: localization standard deviation

One stationary isolated emitter, known Gaussian PSF, independent photons, no background.

Independent photons reduce the centroid variance



Independence removes the covariance terms.

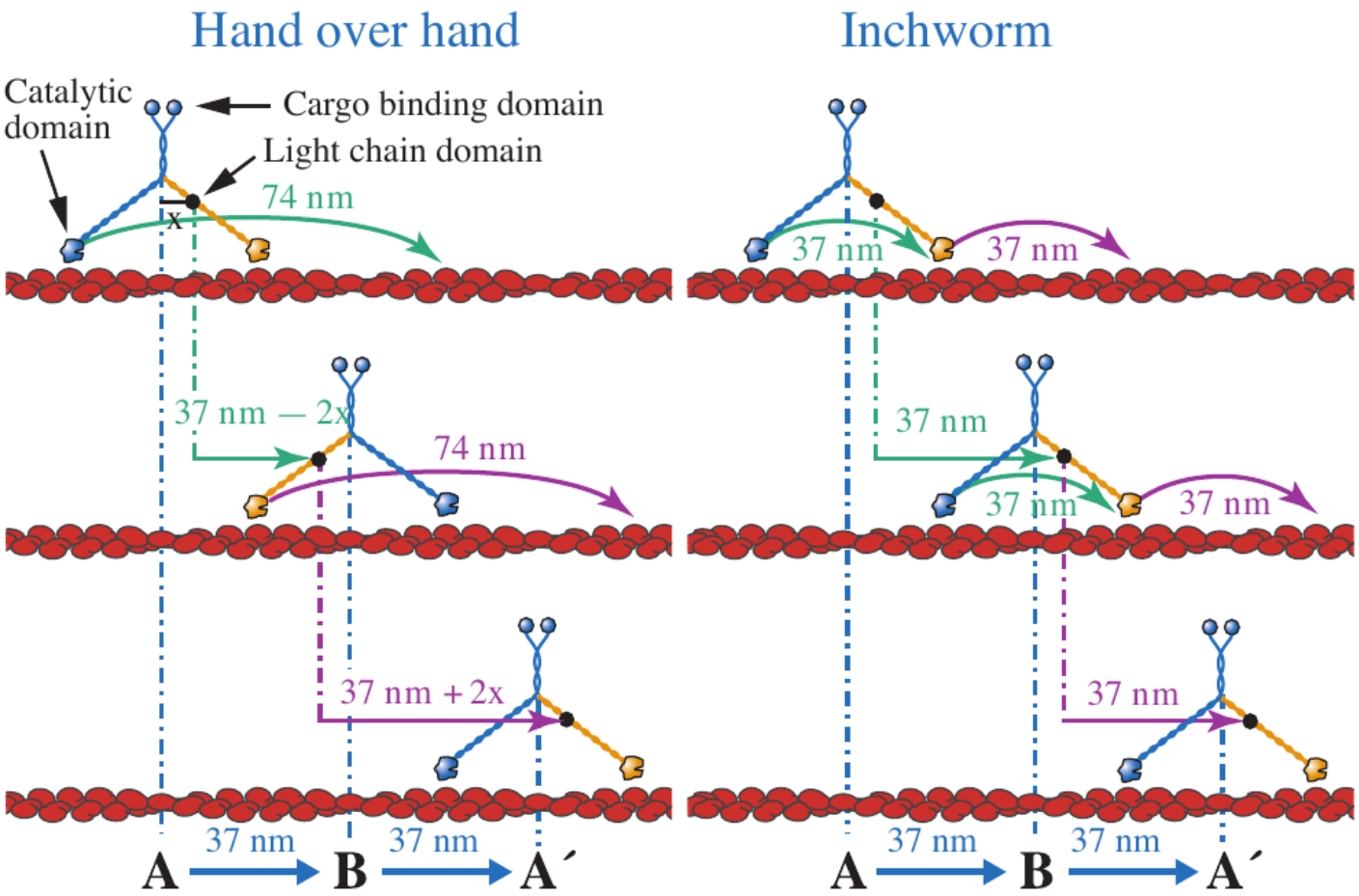
$$\mathrm{Var}(\bar{x}) = \frac{1}{N^2}\sum_{i=1}^{N}\mathrm{Var}(x_i) = \frac{Ns^2}{N^2}$$

$$\sigma_{\mathrm{loc}} = \frac{s}{\sqrt{N}}$$

100× more photons
give 10× smaller σloc.

The displayed spot keeps s = 93 nm. The center intervals use continuous photon coordinates and known s.

FIONA follows a dye on one myosin V lever arm



Two heads bind actin.
ATP powers stepping.

A dye labels calmodulin on one lever arm.

Which part of the motor does the dye report?

Yildiz et al., Science 300, 2061–2065 (2003), Fig. 1. FIONA: fluorescence imaging with one-nanometer accuracy.

The dye alternates between two geometric offsets

Completed steps	Midpoint	Dye offset	Dye position
0	c	−x	c − x
1	c + d	+x	c + d + x
2	c + 2d	−x	c + 2d − x

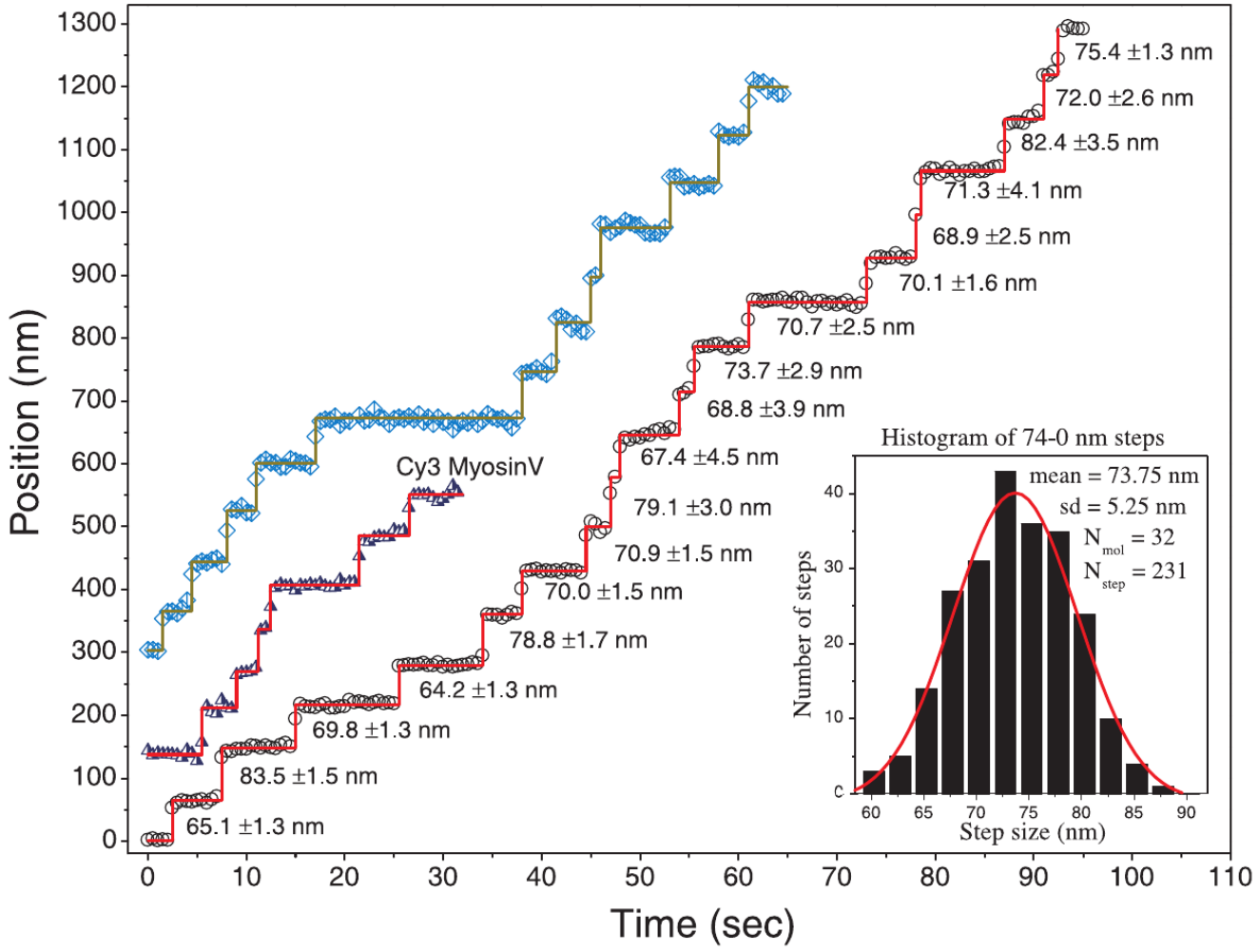
$$\Delta z_1 = (c + d + x) - (c - x) = d + 2x$$

$$\Delta z_2 = (c + 2d - x) - (c + d + x) = d - 2x$$

$$x = \frac{d}{2} \Rightarrow (\Delta z_1, \Delta z_2) = (2d, 0) \simeq (74, 0)\,\mathrm{nm}$$

d ≈ 37 nm. These are successive resting positions projected along the track, under the hand-over-hand geometry.

A lever-arm dye shows ~74-nm visible jumps



Shown at left:
~74-nm visible jumps

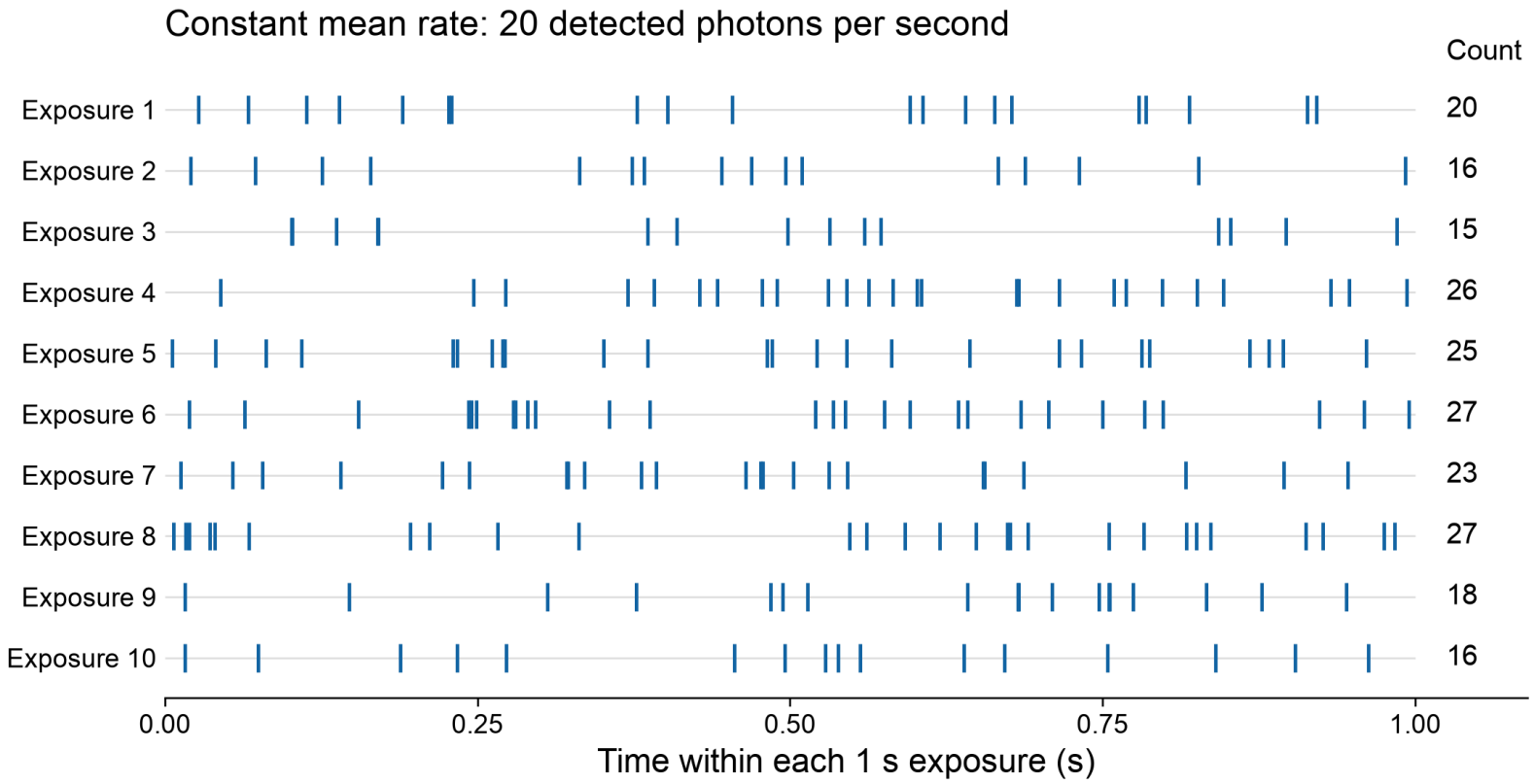
Other label positions reported in the paper:

~52 / 23 nm
~42 / 33 nm

Each alternating pair sums to about 74 nm.

Yildiz et al. (2003), Fig. 3. Other step pairs: Figs. 4–5. Zero-displacement steps remain hidden in the position trace.

Shot noise comes from counting random events



M independent time bins

$X_k = 0$ or 1; $P(X_k = 1) = p$

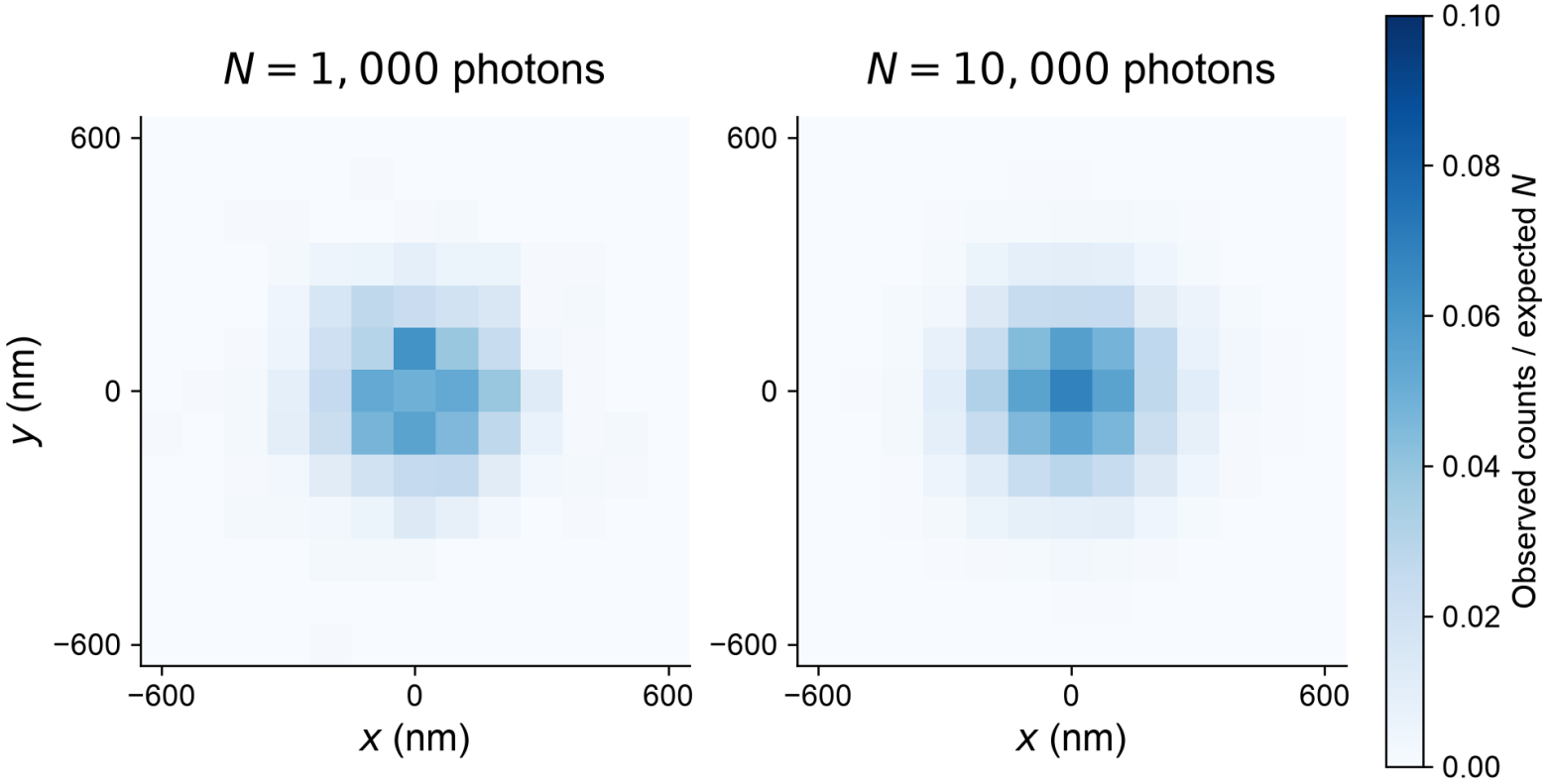
$$Z = \Sigma X_k$$

Mean = Mp

Variance = Mp(1 − p)

Rare-event limit: mean = variance = μ

More photons reduce the relative noise



Same PSF and pixels. The common normalized scale makes relative noise comparable.

$$\mathrm{SD} = \sqrt{\mu}, \qquad \frac{\mathrm{SD}}{\mathrm{mean}} = \frac{1}{\sqrt{\mu}}$$

Central pixel:

68 ± 8 counts

682 ± 26 counts

± denotes one standard deviation. The optical PSF width is the same in both images.

Subtracting background leaves its fluctuations

N: mean signal B: known mean background

Mean signal estimate: $E[Y - B] = N$

Its variance: $\mathrm{Var}(Y - B) = N + B + \sigma_r^2$

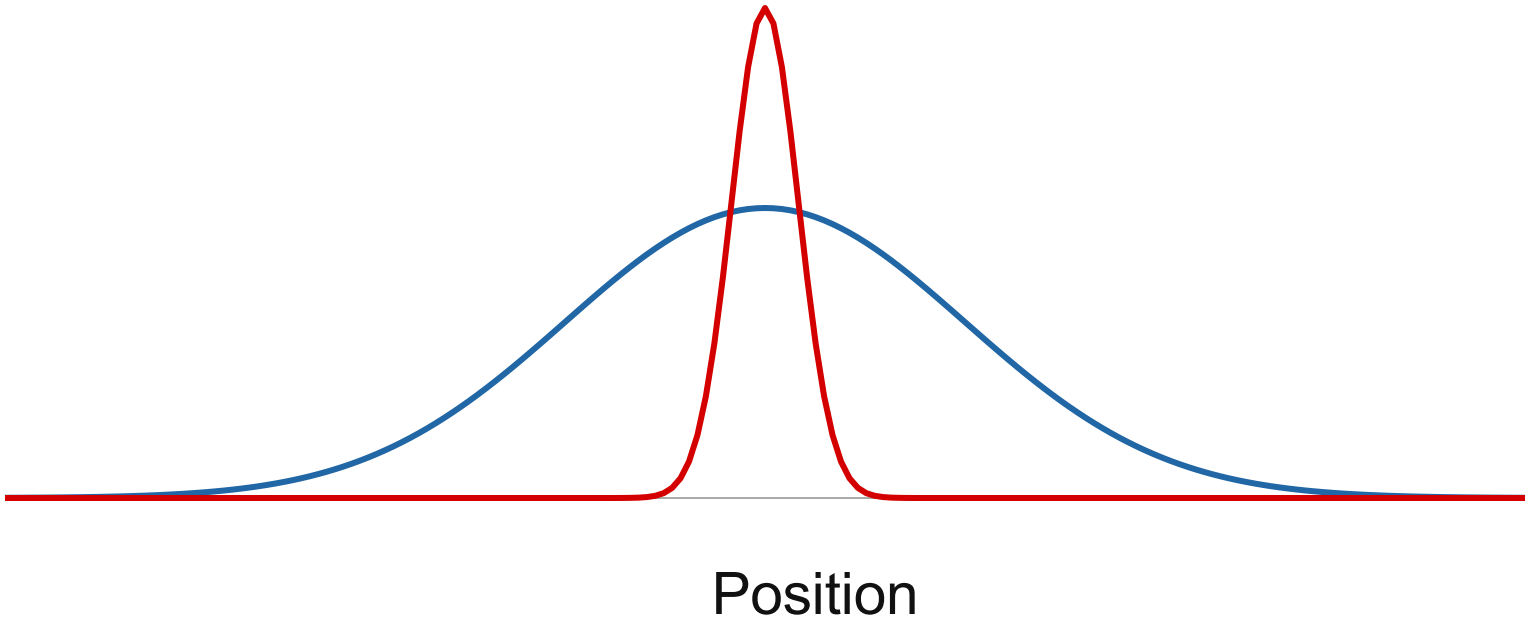
$$\mathrm{SNR} = \frac{N}{\sqrt{N + B + \sigma_r^2}}$$

Example: $N = 200$, $B = 100$, $\sigma_r^2 = 25$ $\rightarrow$ SNR = 11.1

Subtracting a perfectly known mean does not subtract the random background events.

Spot width and uncertainty in its center

Photon positions **Estimated centers**



$$\sigma_{\mathrm{loc}} = \frac{s}{\sqrt{N}}$$

s = 150 nm

N = 1,000: 4.7 nm

A broad spot can still reveal its center precisely.

Ideal isolated Gaussian emitter. Curves use arbitrary vertical scales; center width is illustrative.